

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Meditech AI

Challenge ID: PS-06

Title: Data-Driven Air-Pollution Analysis & Health Advisory System

1. Executive Summary & Problem Overview

Deteriorating ambient air quality is one of the most critical public health emergencies in urban and industrial corridors across India. High concentrations of particulate matter ($PM_{2.5}$, PM_{10}) and toxic gaseous pollutants (NO_2 , SO_2 , CO , O_3) trigger severe acute and chronic respiratory ailments, cardiovascular distress, and premature mortality—disproportionately affecting children, asthmatics, outdoor workers, and the elderly. While extensive government monitoring networks (such as CPCB and SAFAR) and satellite remote sensing systems capture massive volumes of air-quality data, this information remains siloed in complex statistical tables or static AQI numbers. Citizens and municipal healthcare bodies lack predictive insights that forecast pollution spikes before they happen and lack actionable, location-specific, personalized medical advisories.

The objective of **PS-06 (Data-Driven Air-Pollution Analysis & Health Advisory System)** is to build a software-only, data-driven analytics platform that uses publicly available environmental datasets to analyze pollution trends, predict high-pollution windows, visualize spatial air quality levels, and generate personalized, localized health advisories. The solution must rely strictly on existing open datasets or public APIs and must not require any physical hardware, external sensors, or IoT components.

2. Core Personas & Access Roles

1. **Vulnerable Citizens & Patients:** Asthmatics, cardiopulmonary patients, parents, and athletes who receive personalized risk warnings, safest outdoor workout hours, and indoor protection tips.
2. **School Administrators & Event Organizers:** Institutions that access 24–72 hour air quality forecasts to schedule recess, outdoor sports, or cancel physical outdoor assemblies.
3. **Public Health Officers & Epidemiologists:** Healthcare authorities who monitor ward-level pollution correlations with respiratory outpatient trends and evaluate clean-air interventions.

4. **General Public / Community:** Users who access live interactive geospatial air quality maps, pollutant breakdowns, and neighborhood comparative rankings.

3. Key Functional Modules & Technical Deliverables

Module 1: Public Environmental Data Ingestion & Preprocessing

- **Open Dataset Integration:** Ingests live and historical data via public APIs and open repositories (CPCB Open Data, OpenAQ, WAQI, NASA Giovanni / MODIS AOD, Copernicus Atmosphere Data Store, or Open-Meteo Air Quality API).
- **Multi-Pollutant Aggregation:** Processes standard sub-indices for $PM_{2.5}$, PM_{10} , NO_2 , SO_2 , CO , NH_3 , and ground-level O_3 , computing standard National Air Quality Index (NAQI) scores.
- **Meteorological Correlation:** Merges atmospheric variables (wind speed, direction, ambient temperature, relative humidity, boundary layer height) to track thermal inversion and pollutant dispersion.

Module 2: Trend Analysis & Spatial Visualization

- **Spatial Pollution Heatmaps:** Interactive, color-coded geographical maps (Choropleth/Interpolated IDW) visualizing station-level and interpolated neighborhood pollution levels.
- **Historical Trend Mining:** Identifies diurnal patterns, weekday vs. weekend variances, seasonal stubble/festival spikes, and industrial emissions corridors.
- **Station Comparison & Anomaly Alerts:** Benchmarks multiple monitoring stations within a district to detect sudden local pollution surges.

Module 3: Predictive Forecasting Engine (ML/AI)

- **24–72 Hour AQI Forecasting:** Time-series ML model predicting future AQI levels and dominant pollutant categories based on historical trends and meteorological forecasts.
- **Pollution Spike Warning Window:** Flags high-risk time frames (e.g., *"Anticipated severe $PM_{2.5}$ spike between 5:30 AM and 8:30 AM tomorrow due to low wind dispersion"*).
- **Explainable Pollution Drivers:** Highlights primary contributing factors to predicted spikes (e.g., high atmospheric stagnation, vehicular peak hours, humidity trapping).

Module 4: Personalized & Actionable Health Advisory

- **Profile-Based Risk Stratification:** Customizes advisories according to user health profiles (Healthy Adult, Pediatric, Geriatric, Asthmatic/COPD, Cardiovascular patient).
- **Prescriptive Action Guidance:** Delivers concrete recommendations (e.g., optimal N95 mask usage, air purifier run schedules, avoiding morning outdoor jogs, saline nasal gargle reminders).
- **Automated Vernacular Notification Dispatch:** Dispatches automated email, SMS, or web-push advisory digests in English, Odia, and Hindi.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web, React.js (Vite) / Next.js / Tailwind CSS / Leaflet.js / Mapbox GL / Recharts / Chart.js

Backend & APIs, Python (FastAPI / Flask) or Node.js (Express)

Database & Caching, PostgreSQL with PostGIS / TimescaleDB (Time-series) / MongoDB + Redis for API caching

Data Science & ML Models, "Python (Pandas, NumPy, Scikit-learn, XGBoost, Prophet / ARIMA / LSTM for time-series forecasting)"

Open Data Sources, "OpenAQ API, CPCB portal datasets, Open-Meteo Air Quality API, WAQI API, NASA EarthData"

Advisories & NLP, "Google Gemini API / Hugging Face (for health advisory summarization), Web Push / Twilio API"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Data Ingestion & Multi-Pollutant Processing (25 Points):** Comprehensive integration of open air quality data, correct NAQI calculation, and robust preprocessing pipelines without physical sensors.
2. **Predictive Accuracy & Forecasting Logic (25 Points):** Soundness and reliability of the time-series forecasting models in identifying high-pollution windows and trends.
3. **Clinical Utility of Health Advisories (20 Points):** Real-world medical relevance, safety accuracy, and personalization of health advice for vulnerable groups.
4. **Geospatial UI/UX & Data Storytelling (15 Points):** Clarity of spatial heatmaps, responsive design, interactive pollutant breakdown, and intuitive user experience.
5. **Software-Only Architecture & Code Standards (15 Points):** Clean GitHub repository, reproducible mock datasets/APIs, .env.example, and seamless live deployment.

6. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository with structured commits, .env.example, sample data scripts, and clean README setup documentation.
2. **Live Deployment:** Hosted interactive web application (Vercel / Render / Netlify / AWS).
3. **Demo Video / Deck:** A 3-minute video showing:
 4. Ingestion of open air-quality datasets.
 5. Spatial map visualization and historical trend analysis.
 6. Running the ML forecasting engine for future pollution spikes.
 7. Generating a personalized, vernacular health advisory for high-risk demographics.